

Srizoni Maity

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PERSONAL PROFILE

A self-motivated and passionate individual seeking to pursue a research-integrated Master's degree in Computer Science, with a strong interest in Explainable AI, Multimodal AI, Trustworthy Machine Learning, Computer Vision, and NLP. Driven to develop interpretable, robust, and scalable intelligent systems through rigorous research and experimentation, with the long-term goal of contributing impactful solutions to emerging challenges in Artificial Intelligence.

EDUCATION

The Neotia University | West Bengal, India

Date: 2022-2026

**Bachelor of Technology (B.Tech.) in Computer Science and Engineering,
Specialization in Data Science**

GPA: 8.17/10

Capstone Project: "Nova: Multimodal AI Stress Intelligence Platform" (Supervisor: Prof. Bera)

RESEARCH INTERESTS

My research interests span Explainable Artificial Intelligence (XAI), Multimodal Artificial Intelligence, Computer Vision, Natural Language Processing, Trustworthy Machine Learning, and Edge Intelligence with a goal of advancing interpretable, robust, and scalable Intelligence systems for reliable real-world decision-making.

RESEARCH EXPERIENCE

Research Intern | Indian Institute of Information Technology Guwahati
(Research Supervisor: Dr. Masihuddin)

Date: Jan-Jun 2026

- Proposed and validated a leakage-free XAI framework for early PCOS prediction using survey data, integrating feature engineering, consensus feature selection, nested cross-validation, and SHAP explainability.
- Experimentally evaluated multiple machine learning models, with the optimized CatBoost model achieving 0.9138 ROC-AUC with robust evaluation through calibration and uncertainty quantification.

INDUSTRY EXPERIENCES

Project Intern | Defence Research and Development Organization (DRDO)

Date: Aug-Dec 2025

Optimized mission-critical radar software through compiler-level optimization, low-level systems tuning, and performance profiling while developing a real-time radar visualization and analytics framework using Python and MATLAB, reducing execution time from 3.66 s to 1.57 s (57.1%) and enabling high-throughput sensor processing, faster situational awareness, and improved operational decision-making; performance evaluation findings contributed to DRDO internal R&D documentation.

Project Intern | AILABS, DataCore

Date: Jun-Jul 2025

Engineered an end-to-end MLOps pipeline for retail sales forecasting using Prophet, XGBoost, and LSTM, integrating automated preprocessing, reproducible training, and MLflow-based experiment tracking. Deployed a Flask-Plotly analytics platform for real-time forecasting, inventory intelligence, and scalable decision support.

Project Intern | West Bengal Electronics Industry Development Corporation Limited Date: Jun-Aug 2024
Developed a machine learning-driven academic analytics platform integrating Streamlit and Plotly for interactive CGPA prediction, student performance analytics, and real-time educational insights, enabling data-driven academic monitoring and decision support.

RESEARCH CONTRIBUTIONS

Manuscripts Under Preparation

An Explainable Machine Learning Framework for Early Prediction of Polycystic Ovary Syndrome (PCOS) Using Dietary, Lifestyle, and Industrial Trans-Fat Consumption Among College-Age Women

Open Research Dataset

DR-Multimodal India ([Kaggle](#))

Publicly released multimodal healthcare dataset supporting AI-driven medical research.

PROJECT WORK

Nova: Multimodal AI Stress Intelligence Platform (Capstone Project)

- Designed a research-grade multimodal AI architecture integrating EfficientNet-ViT, Wav2Vec2-HuBERT, and RoBERTa to jointly model visual, acoustic, linguistic, physiological, and psychometric representations for stress assessment.
- Introduced an Adaptive Decision Fusion framework with confidence-weighted multimodal reasoning to aggregate heterogeneous AI predictions into a unified stress inference pipeline.
- Delivered an end-to-end explainable AI platform achieving 96.3% classification accuracy, translating multimodal predictions into personalized mental-health recommendations.

NeuraTune: Emotion-Aware Music Recommendation System

- Architected a CNN-based affective computing pipeline for real-time facial emotion recognition and personalized music recommendation.
- Engineered an end-to-end computer vision-driven recommendation system enabling adaptive, context-aware user experiences.

EdgeVision: TinyML-Based Privacy-Preserving Anomaly Detection for Smart Cities

- Architected a TinyML-powered edge vision framework for real-time anomaly detection, enabling privacy-preserving on-device inference for smart city IoT applications.
- Optimized lightweight deep learning models for resource-constrained edge devices, reducing cloud dependency while enabling low-latency intelligent surveillance.

Bhasha Arogya: Cross-Language AI Healthcare Chatbot

- Architected a RAG-enhanced multilingual LLM framework for cross-language medical query understanding and context-grounded healthcare assistance.
- Integrated semantic retrieval, translation-aware reasoning, and multilingual response generation into an end-to-end healthcare intelligence pipeline.

TECHNICAL SKILLS

Programming: Python, SQL, C/C++

AI Research: Machine Learning • Deep Learning • Computer Vision • Natural Language Processing • Large Language Models (LLMs) • Multimodal AI • Explainable AI (XAI) • Affective Computing • Speech Emotion Recognition • Edge AI • TinyML • Privacy-Preserving AI

Research & Modeling: Representation Learning • Statistical Learning • Feature Engineering • Feature Selection • Hyperparameter Optimization • Nested Cross-Validation • Model Calibration • Decision Fusion • Model Interpretability (SHAP) • Uncertainty Quantification

Frameworks: PyTorch • TensorFlow • TensorFlow Lite • Hugging Face Transformers • Scikit-learn • OpenCV • XGBoost • CatBoost • NumPy • Pandas

Development Tools: Git • GitHub • Jupyter Notebook • VS Code • Streamlit

WORKSHOPS & SEMINARS

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| • National Level Seminar on “From Discovery to Conservation: Scientific Approaches to Coastal Bioresource Management” — Egra, West Bengal | Date: 2025 |
| • ML/DL Model Deployment in Healthcare for Business Applications using FastAPI — 20-hour hands-on workshop | Date: 2025 |
| • Data Modelling using Advanced Java and Oracle — 2-day hands-on workshop | Date: 2025 |
| • Data Visualization Techniques in Business Analytics using Tableau/Power BI (24h) | Date: 2024 |

ACHIEVEMENTS

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| • Tejashee Merit Scholarship The Neotia University | Date: 2022-2026 |
| • Winner, National Science Day Poster Presentation | Date: 2023 |
| • Swami Vivekananda Merit-cum-Means Scholarship Government of West Bengal | Date: 2022,2023 |

CERTIFICATIONS & TRAININGS

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| • Elements of AI – University of Helsinki | Date: 2025 |
| • Introduction to Generative AI – Google Cloud / Simplilearn | Date: 2025 |
| • Machine Learning with Python – freeCodeCamp (300h) | Date: 2024 |

COMMUNITY ENGAGEMENT & OUTREACH

Digital Literacy Outreach Initiative | Diamond Harbour, West Bengal, India Date: 2024
Facilitated foundational computer literacy training for students from underserved rural communities.

International Coastal Clean-up Day | Digha Beach, West Bengal, India Date: 2023
Contributed to a community-based coastal conservation and environmental stewardship initiative.

REFERENCES

Dr. Usha Rani Gogoi | ✉ usharani.gogoi@tnu.in
Head of the Dept. of Computer Science & Engineering | The Neotia University

Dr. Ayan Chatterjee | ✉ ayan.chatterjee@tnu.in
Assistant Professor, Dept. of Basic Science | The Neotia University